

National Coverage Determination (NCD)

Next Generation Sequencing (NGS)

90.2

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Tracking Information

Publication Number

100-3

Manual Section Number

90.2

Manual Section Title

Next Generation Sequencing (NGS)

Version Number

1

Effective Date of this Version

03/16/2018

Ending Effective Date of this Version

01/27/2020

Implementation Date

04/08/2019

Description Information

Benefit Category

Diagnostic Laboratory Tests

Diagnostic Services in Outpatient Hospital

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Clinical laboratory diagnostic tests can include tests that, for example, predict the risk associated with one or more genetic variations. In addition, in vitro companion diagnostic laboratory tests provide a report of test results of genetic variations and are essential for the safe and effective use of a corresponding therapeutic product. Next Generation Sequencing (NGS) is one technique that can measure one or more genetic variations as a laboratory diagnostic test, such as when used as a companion in vitro diagnostic test.

Patients with cancer can have recurrent, relapsed, refractory, metastatic, and/or advanced stages III or IV of cancer. Clinical studies show that genetic variations in a patient's cancer can, in concert with clinical factors, predict how each individual responds to specific treatments.

In application, a report of results of a diagnostic laboratory test using NGS (i.e., information on the cancer's genetic variations) can contribute to predicting a patient's response to a given drug: good, bad, or none at all. Applications of NGS to predict a patient's response to treatment occurs ideally prior to initiation of such treatment.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on or after March 16, 2018, the Centers for Medicare & Medicaid Services (CMS) has determined that Next Generation Sequencing (NGS) as a diagnostic laboratory test is reasonable and necessary and covered nationally, when performed in a Clinical Laboratory Improvement Amendments (CLIA)-certified laboratory, when ordered by a treating physician, and when all of the following requirements are met:

1. Patient has:

- either recurrent, relapsed, refractory, metastatic, or advanced stage III or IV cancer; and,
- either not been previously tested using the same NGS test for the same primary diagnosis of cancer, or repeat testing using the same NGS test only when a new primary cancer diagnosis is made by the treating physician; and,
- decided to seek further cancer treatment (e.g., therapeutic chemotherapy).

2. The diagnostic laboratory test using NGS must have:

- Food & Drug Administration (FDA) approval or clearance as a companion in vitro diagnostic; and,
- an FDA-approved or -cleared indication for use in that patient's cancer; and,

- results provided to the treating physician for management of the patient using a report template to specify treatment options.

C. Nationally Non-Covered

Effective for services performed on or after March 16, 2018, NGS as a diagnostic laboratory test for patients with cancer are non-covered if the cancer patient does not meet the criteria noted in section B.1. above.

D. Other

1. Effective for services performed on or after March 16, 2018, Medicare Administrative Contractors (MACs) may determine coverage of other NGS as a diagnostic laboratory test for patients with cancer only when the test is performed in a CLIA-certified laboratory, ordered by a treating physician, and the patient has:
 - either recurrent, relapsed, refractory, metastatic, or advanced stages III or IV cancer; and,
 - either not been previously tested using the same NGS test for the same primary diagnosis of cancer or repeat testing using the same NGS test was performed only when a new primary cancer diagnosis is made by the treating physician; and,
 - decided to seek further cancer treatment (e.g., therapeutic chemotherapy).

(This NCD last reviewed March 2018.)

Transmittal Information

Transmittal Number

215

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/2019Downloads/R215NCD.pdf> 

Revision History

07/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10261](#) ) (CR11905)

06/2020 - Transmittal 2439, dated February 21, 2020, is being rescinded and replaced by Transmittal 10193, dated, June 19, 2020 remove Current Procedural Technology (CPT) code 0048U from business requirement 11655.1 and corresponding removals of CPT 0048U and

its associated diagnosis codes from the National Coverage Determination (NCD) 90.2 Next Generation Sequencing (NGS) spreadsheet. This revision is necessary because the CPT code does not meet the policy criteria in NCD 90.2 for NGS. All other information remains the same. ([TN 10193](#)) (CR11655)

05/2020 This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 10092](#)) (CR11749)

02/2020 This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 2439](#)) (CR11655)

04/2019 - Transmittal 214, dated March 6, 2019, is being rescinded and replaced by Transmittal 215, dated, April 10, 2019 to incorporate missing CPT codes on the diagnosis code attachments. All other information remains the same. ([TN 215](#)) (CR10878)

03/2019 - Transmittal 210, dated November 30, 2018, is being rescinded and replaced by Transmittal 214, dated, March 6, 2019 to extend the implementation date 30 days and update the attached diagnosis code list. This instruction is being re-communicated to include attachment that was omitted. The Transmittal number, Dated issued and all other information remain the same. ([TN 214](#)) (CR10878)

11/2018 - The purpose of this Change Request (CR) is to inform contractors that effective March 16, 2018, the Centers for Medicare & Medicaid Services covers diagnostic laboratory tests using next generation sequencing when performed in a Clinical Laboratory

Improvement Amendments- certified laboratory when ordered by a treating physician and when specific requirements are met. THIS CHANGE REQUEST (CR) AND PUBLICATION (PUB.) 100-03 MANUAL TRANSMITTAL REFLECTS THE AGENCY’S FINAL DECISION DATED MARCH 16, 2018, REGARDING THE NATIONAL COVERAGE DETERMINATION (NCD) 90.2, ON NEXT GENERATION SEQUENCING (NGS). A SUBSEQUENT CR WILL BE RELEASED AT A LATER DATE THAT CONTAINS A PUB. 100-04 CLAIMS PROCESSING MANUAL UPDATE AND FURTHER, ACCOMPANYING INSTRUCTIONS. UNTIL THAT TIME, THE MEDICARE ADMINISTRATIVE CONTRACTORS (MACS) SHALL BE RESPONSIBLE FOR IMPLEMENTING NCD 90.2.([TN 210](#)) (CR10878)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Next Generation Sequencing \(NGS\) for Medicare Beneficiaries with Advanced Cancer \(CAG-00450N\)](#)
- [First reconsideration for Next Generation Sequencing \(NGS\) for Medicare Beneficiaries with Advanced Cancer \(CAG-00450R\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Next Generation Sequencing (NGS)	2	01/27/2020 - N/A	View
Next Generation Sequencing (NGS)	1	03/16/2018 - 01/27/2020	You are here